IT Risk Assessment
AuroraTel Communications

Risk Number/Identifier	R-001
Risk Name	5G Core / IMS Network Compromise
Assessment Status	ACTIVE
Date of Risk Assessment	Jul, 2026

Risk Description

A successful compromise of AuroraTel's 5G Core, 4G EPC, or IMS voice platform could disrupt voice and data services on a national scale. As critical national telecom infrastructure, these systems are of interest to sophisticated threat actors, including nation-state groups, particularly given AuroraTel's rapid 5G rollout across urban centers and industrial corridors.

Risk Identification and Assessment Approach

Risk information was gathered through consultation with the Network Engineering and SOC teams, and by reviewing industry intelligence on attacks against telecom core infrastructure. The scope of this assessment covers the 5G Core, 4G EPC, and IMS voice platforms. Assessment prepared by the Chief Information Security Officer (CISO).

Key Controls

Control Strength: Weaker

- Network segmentation between the core network and OSS/BSS systems.
- IAM/MFA for core network administrative access.
- SIEM correlation rules covering core network events.
- Vendor-supplied security patches for core network elements.

Gaps/Weaknesses

AuroraTel's controls provide a reasonable baseline but have notable weaknesses given the criticality of the assets involved. Some notable control gaps include:

- Legacy 4G EPC interfaces have a limited patch cadence.
- Some OSS tools remain on flat network zones with the core, weakening segmentation.
- No dedicated 24/7 threat-hunting capability focused on core network systems.

Risk Evaluation

Frequency Rationale: Given the high value of telecom core infrastructure as a target, a serious compromise attempt is plausible roughly once every three to five years, though monitoring gaps mean detection could be delayed if an attempt succeeded.
Impact Rationale: A limited compromise may be contained to a single network element; a severe event could cause a nationwide voice/data outage, trigger regulatory penalties, and significantly damage AuroraTel's brand and national security standing.

Business Areas Impacted:

- All subscribers nationwide
- Regulatory Compliance (telecom regulator)
- National security obligations
- Brand / Reputation

Summary: The overall risk exposure rating reflects the low likelihood but severe potential impact typical of critical-infrastructure risk. Segmentation and threat-hunting improvements would meaningfully reduce the probability of an undetected compromise.

Risk Exposure		
Control Strength	Weaker	Legacy interfaces and incomplete segmentation between core and OSS/BSS
Frequency	Occasional	Once every three to five years
Most Likely Annual Loss	Low-Moderate	PKR 80M per year
One-time Maximum Loss	Severe	PKR 3,000M per event

Risk Response

The following actions are proposed to bring this risk within tolerance:

- Complete the network segmentation project separating the core network from OSS/BSS (Owner: CTO; Target: 9 months).
- Establish 24/7 core-network threat hunting within the SOC (Owner: CISO; Target: 6 months).